

Lab 27: NumPy Array Creation and Basic Operations

Aim

To create **NumPy arrays using different methods** and perform basic mathematical operations such as arithmetic operations, dot product, and mathematical functions.

Algorithm

1. Import the NumPy library.
2. Create arrays using `array()`, `zeros()`, `ones()`, `arange()`, and `linspace()`.
3. Perform element-wise arithmetic operations.
4. Calculate the dot product of two arrays.
5. Apply mathematical functions to the arrays.
6. Display the results.

Program

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```
import numpy as np
```

```
# Creating arrays
```

```
array1 = np.array([1, 2, 3, 4, 5])
```

```
zeros_array = np.zeros(5)
```

```
ones_array = np.ones(5)
```

```
range_array = np.arange(1, 10, 2)
```

```
linear_array = np.linspace(0, 10, 5)
```

```

print("Original Array:", array1)
print("Zeros Array:", zeros_array)
print("Ones Array:", ones_array)
print("Arange Array:", range_array)
print("Linspace Array:", linear_array)

# Basic arithmetic operations
array2 = np.array([5, 4, 3, 2, 1])

print("\nAddition:", array1 + array2)
print("Subtraction:", array1 - array2)
print("Multiplication:", array1 * array2)

# Dot product
print("\nDot Product:", np.dot(array1, array2))

# Mathematical functions
print("Square Root:", np.sqrt(array1))
print("Square:", np.square(array1))

```

Sample Output

```

Original Array: [1 2 3 4 5]
Zeros Array: [0. 0. 0. 0. 0.]
Ones Array: [1. 1. 1. 1. 1.]
Arange Array: [1 3 5 7 9]
Linspace Array: [ 0.  2.5  5.  7.5 10. ]

```

```

Addition: [6 6 6 6 6]
Subtraction: [-4 -2  0  2  4]
Multiplication: [5 8 9 8 5]

```

Dot Product: 35

Square Root: [1. 1.41421356 1.73205081 2. 2.23606798]

Square: [1 4 9 16 25]

Result

Thus, **NumPy arrays were successfully created using different methods**, and basic arithmetic, dot product, and mathematical operations were performed.